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| | **GLOBAL ECONOMICS**<br>**GRADE: 10TH**<br>**CATCH-UP ACTIVITY**<br>**ANALYSIS OF DECISIONS**<br>**TEACHER'S NAME: Nicolás López Cuéllar** | **FIRST TERM**<br>**2025–2026** |

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| **Learning objective:**<br>Interpret decision alternatives, events, consequences, and states of nature in applied decision-making scenarios. | **Assessment criteria:**<br>C2: Interprets decision alternatives, events, consequences, and states of nature. |

## Criteria assessment

This activity evaluates criterion C2. Read each problem and complete the required decision-analysis tasks.

## 1. Problem description

A bottled tea company must choose one of two monthly launch options for a new flavor line: Option A (premium glass packaging) or Option B (standard recyclable packaging). Actual demand for the month is uncertain and can be high or low. Estimated net profits (thousand USD) are already available: A gives 98 under high demand and 20 under low demand; B gives 80 under high demand and 37 under low demand.

## 2. Problem description

A regional home-goods wholesaler must choose one of three stocking policies for a holiday campaign: Policy A (premium assortment), Policy B (balanced assortment), or Policy C (value assortment). Demand can be strong, moderate, or weak. Net profits (thousand USD) are estimated as follows: A: 142, 90, 14; B: 124, 96, 40; C: 107, 78, 50.

## 3. Problem description

A sportswear producer must choose one of two monthly production mixes: Mix A (performance line) or Mix B (basic line). Demand may be very high, high, medium, or low. Profit equals units sold times contribution per unit. Contribution per unit is \$19 for Mix A and \$15 for Mix B. Expected units sold are: A: 8,800 (very high), 7,100 (high), 4,900 (medium), 2,500 (low); B: 10,000, 8,100, 5,900, 3,300 respectively.

## 4. Problem description

A digital news publisher must select one of three distribution plans for next month: Plan A (direct subscriptions), Plan B (platform partnership), or Plan C (hybrid channel). Demand can be high or low. Profit model: Profit = Fixed component + (Subscribers × Contribution per subscriber). Parameters: A: fixed \$25,000, contribution \$6.4; B: fixed \$16,000, contribution \$5.9; C: fixed \$20,000, contribution \$6.0. Subscribers by state: High — A 17,500, B 20,800, C 19,200; Low — A 9,200, B 11,700, C 10,500.

## 5. Problem description

A packaging supplier must choose between two contract structures for a seasonal client: Contract A or Contract B. Demand may be strong, normal, or weak. Contract A profits are given directly (thousand USD): 114, 76, 28. Contract B uses $\text{Profit}_B = 22{,}000 + (\text{Units sold} \times 4.6)$, with units sold 17,500 (strong), 11,800 (normal), 7,200 (weak).

## 6. Problem description

An urban courier company must select one of two fleet plans for a quarterly delivery contract: Fleet A or Fleet B. Revenue can be high or low; operating cost can be low, medium, or high. All values are given directly in thousand USD. Revenue: A = 430 (high), 295 (low); B = 458 (high), 305 (low). Costs: A = 175, 210, 246 for low/medium/high cost; B = 188, 224, 264 for low/medium/high cost. Profit equals revenue minus cost.

## 7. Problem description

A corporate meal-service provider must choose one of three service plans: A, B, or C. Meal demand can be high or low, while input costs can be low or high. Revenue is computed from quantity and price; costs are given directly. Revenue inputs: A: 13,800 meals (high), 8,900 (low), price \$12.2; B: 15,200, 10,300, price \$11.6; C: 13,000, 8,600, price \$12.5. Cost data (thousand USD): A: 103 (low cost), 129 (high cost); B: 109, 136; C: 99, 125. Profit = revenue − cost.

## 8. Problem description

A vocational training operator must choose between Program A and Program B. Tuition revenue is given directly for enrollment-demand states high, medium, and low. Costs depend on instructor rates (low-cost or high-cost state) and must be computed from participant counts. Revenue (thousand USD): A: 365, 285, 212; B: 395, 305, 233. Cost model: Cost = Participants × Unit cost. A: 980 participants (low-cost state) and 1,280 (high-cost state), unit cost \$122. B: 1,040 and 1,340, unit cost \$126. Profit = revenue − cost.

## 9. Problem description

A smart-appliance firm must choose one of three release configurations: Configuration P, Q, or R. Both revenue and cost include fixed and variable components. Demand can be high, medium, or low; operating condition can be normal or disrupted. Revenue models (USD): $R_P = 148{,}000 + 52q$, $R_Q = 132{,}000 + 55q$, $R_R = 144{,}000 + 53q$. Demand quantities: P: 8,700, 6,900, 5,000; Q: 8,500, 6,800, 5,200; R: 8,800, 6,950, 5,100. Cost models (USD): $C_P = 121{,}000 + 27n$, $C_Q = 106{,}000 + 29n$, $C_R = 113{,}000 + 28n$. Cost activity quantities: P: 8,100 (normal), 9,000 (disrupted); Q: 7,900, 8,900; R: 8,000, 9,000.

## 10. Problem description

A specialty coffee chain must decide between two expansion formats for a district rollout: Format A (large stores) or Format B (compact stores). Demand can be high or low, and ingredient costs can be stable or volatile. Projected net profits are already estimated in thousand USD: Format A: 210 (high, stable), 184 (high, volatile), 74 (low, stable), 48 (low, volatile). Format B: 198 (high, stable), 170 (high, volatile), 92 (low, stable), 64 (low, volatile).